

PYTHON PROGRAMMING LAB

I B.TECH - II SEMESTER								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS04	ESC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. To understand the problem-solving approaches.
2. To learn the basic programming constructs in Python.
3. To practice various computing strategies for Python-based solutions to real world problems.
4. To use Python data structures – lists, tuples, dictionaries.
5. To do input/output with files in Python

COURSE OUTCOMES

Upon successful completion of this course, student will be able to :

1. Write, test, and debug simple Python programs.
2. Implement Python programs with conditions and loops.
3. Develop Python programs step-wise by defining functions and calling them.
4. Use Python lists, tuples, dictionaries for representing compound data.
5. Read and write data from/to files in Python

LIST OF PROGRAMS

WEEK - 1	
	a) Write a program to perform different Arithmetic Operations on numbers in Python b) Write a Python program which accepts the radius of a circle from the user and compute the area. c) Write a Python program to get the Python version you are using. d) Write a Python program that accepts an integer (n) and computes the value of n+nn+nnn
WEEK - 2	
	a) Write a Python program to convert temperatures to and from Celsius, Fahrenheit. [Formula: $c/5 = f-32/9$] b) Write a python script to print the current date in the following format "Sun May 29 02:26:23 IST 2017" c) A library charges a fine for every book returned late. For first 6 days the fine is 50 paise, for 10-15 days fine is one rupee and above 15 days fine is 5 rupees. If you return the book after 30 days your membership will be cancelled. d) Write a python program to accept the number of days the member is late to return the book and display the fine or the appropriate message.
WEEK - 3	
	a) Write a python function to find largest of three numbers. b) Write a Python function that prints prime numbers in between 50 and 100. c) Write a python program to find factorial of a number using Recursion. d) Write a function that receives marks received by a student in 6 subjects and returns the average and percentage of these marks. Call this function from main() and print the result in main

WEEK - 4	
Write a program to demonstrate working with Strings and string operations.	
WEEK - 5	
Write a program to demonstrate working with dictionaries in python	
WEEK - 6	
Write a program to demonstrate working with tuples and List in python.	
WEEK - 7	
a) Write a script named hellow.py. This script should prompt the user for the names of two text files. The contents of the first file should be input and written to the second file. b) Write a program that inputs a text file. The program should print all of the unique words in the file in alphabetical order	
WEEK - 8	
Write python programs to demonstrate class & object, static and instance method implementation.	
WEEK - 9	
Write python programs to demonstrate Inheritance and Polymorphism	
WEEK - 10	
Write python programs to demonstrate Exception Handling in python	
WEEK - 11	
Write python programs to demonstrate NumPy library and supporting functions.	
WEEK - 12	
a) Draw an Olympic Symbol in Python using Turtle b) Develop a simple login page using GUI Tkinter	
TEXT BOOKS	
1. Allen B. Downey, Think Python: How to Think like a Computer Scientist, 2 nd Edition, Oâ€™Reilly Publishers, 2016. 2. Karl Beecher, Computational Thinking: A Beginners Guide to Problem Solving and Programming, 1st Edition, BCS Learning and Development Limited, 2017.	

REFERENCE REFERENCES

1. Paul Deitel and Harvey Deitel, Python for Programmers, Pearson Education, 1st Edition, 2021.
2. G Venkatesh and Madhavan Mukund, Computational Thinking: A Primer for Programmers and Data Scientists, 1st Edition, Notion Press, 2021.
3. John V Guttag, Introduction to Computation and Programming Using Python: With Applications to Computational Modeling and Understanding Data Structures, Third Edition, MIT Press, 2021
4. Eric Matthes, Python Crash Course, A Hands – on Project Based Introduction to Programming, 2nd Edition, No Starch Press, 2019. <https://www.python.org/>
5. Martin C. Brown, Python: The Complete Reference, 4th Edition, Mc-Graw Hill, 2018.